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Use Case Diagram

A **use case diagram** is a visual representation used in software engineering to depict the interactions between **system actors** and the **system itself**. It captures the dynamic behavior of a system by illustrating its **use cases** and the roles that interact with them. These diagrams are essential in specifying the system's **functional requirements** and understanding how users will interact with the system. By providing a high-level view, use case diagrams help stakeholders understand the system's functionality and its potential value.

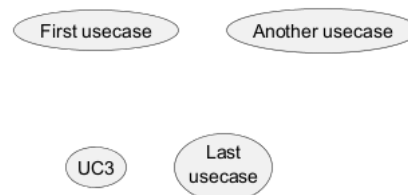
PlantUML offers a unique approach to creating use case diagrams through its text-based language. One of the primary advantages of using PlantUML is its **simplicity and efficiency**. Instead of manually drawing shapes and connections, users can define their diagrams using intuitive and concise textual descriptions. This not only speeds up the diagram creation process but also ensures **consistency and accuracy**. The ability to integrate with various documentation platforms and its wide range of supported output formats make PlantUML a versatile tool for both developers and non-developers. Lastly, being **open-source**, PlantUML boasts a **strong community** that continually contributes to its improvement and offers a wealth of resources for users at all levels.

Usecases

Use cases are enclosed using between parentheses (because two parentheses looks like an oval).

You can also use the `usecase` keyword to define a usecase. And you can define an alias, using the `as` keyword. This alias will be used later, when defining relations.

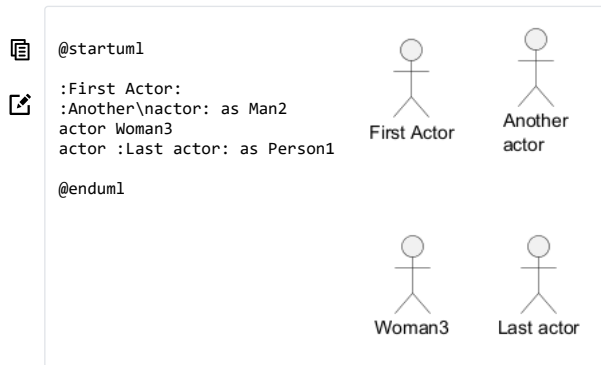
```
@startuml
  (First usecase)
  (Another usecase) as UC2
  usecase UC3
  usecase (Last\nusecase) as UC4
@enduml
```



Actors

The name defining an actor is enclosed between colons.

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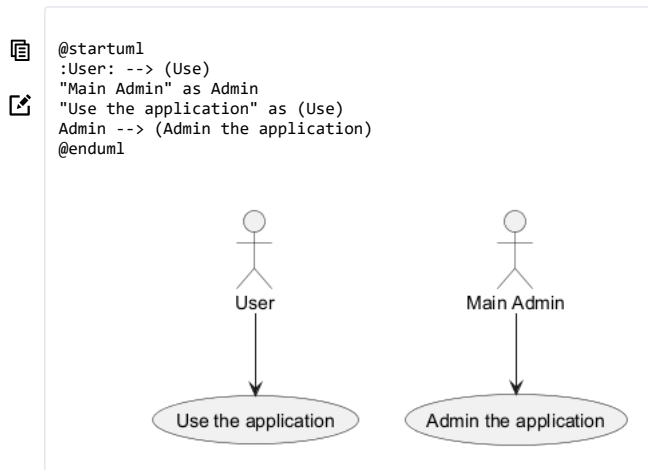


Change Actor style

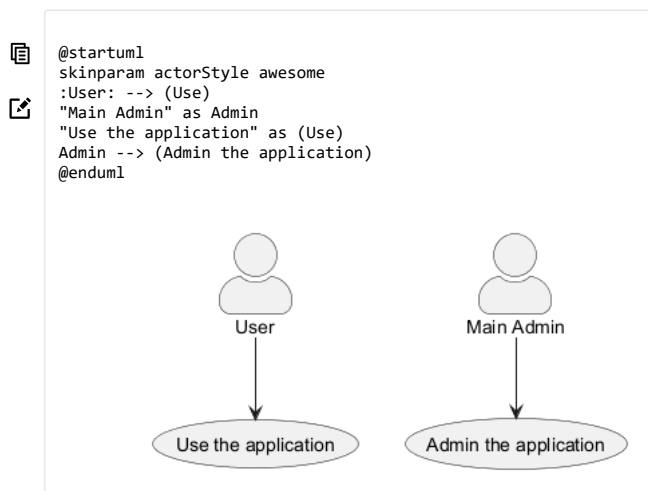
You can change the actor style from stick man (*by default*) to:

- an awesome man with the `skinparam actorStyle awesome` command;
- a hollow man with the `skinparam actorStyle hollow` command.

Stick man (*by default*)

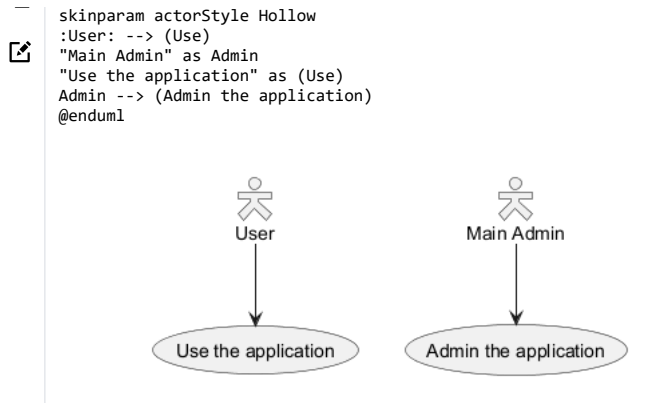


Awesome man



[Ref. [QA-10493](#)]

Hollow man



[Ref. [PR#396](#)]

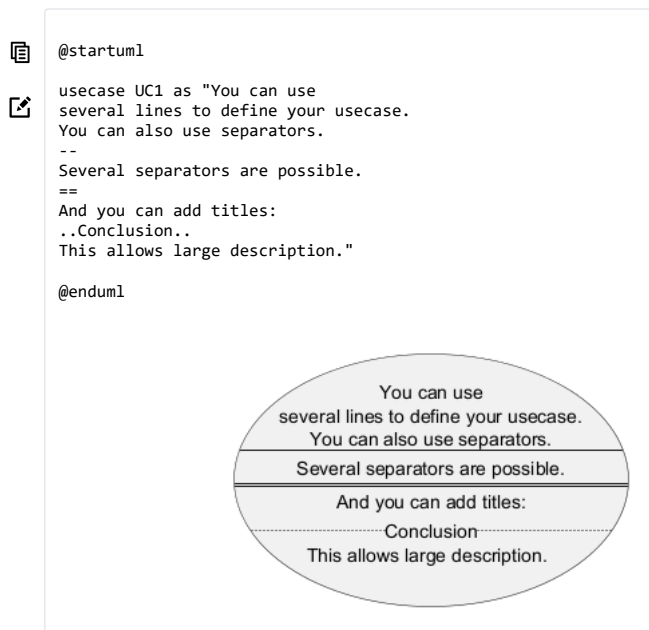
Use cases description

If you want to have a description spanning several lines, you can use quotes.

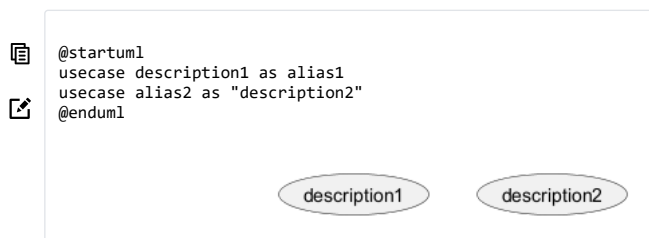
You can also use the following separators:

- -- (dashes)
- .. (periods)
- == (equals)
- _ (underscores)

By using them pairwise and enclosing text between them, you can created separators with titles.



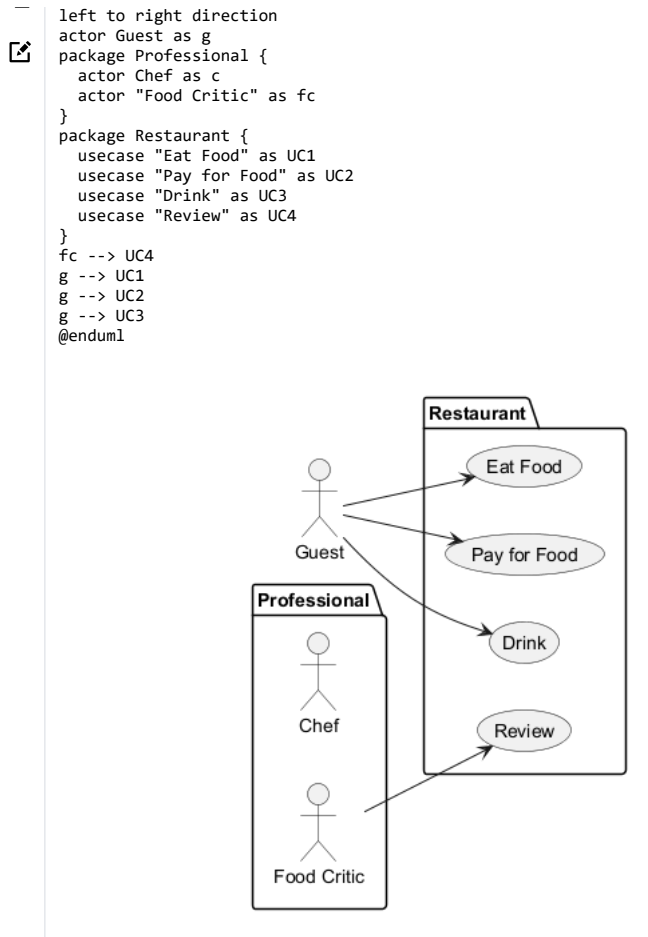
Please note that the alias and the description are switched around from the basic example:



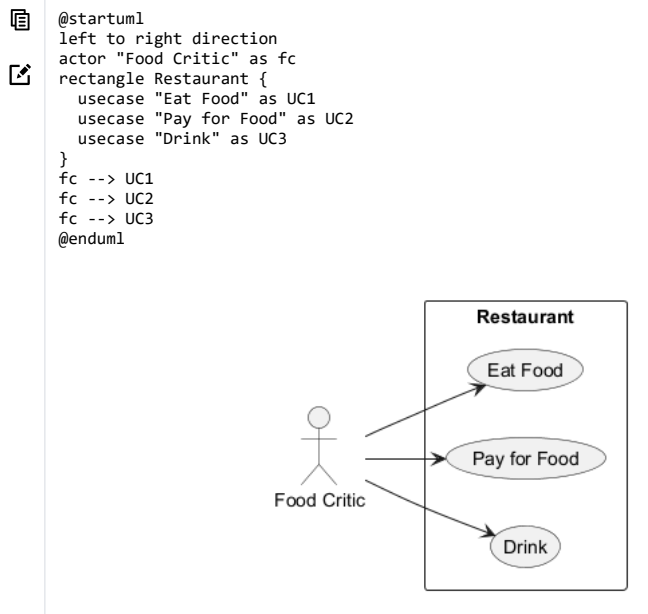
Use package

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You can use `rectangle` to change the display of the package.



Basic example

To link actors and use cases, the arrow `-->` is used.

The more dashes `-` in the arrow, the longer the arrow. You can add a label on the arrow, by adding a `:` character in the arrow definition.

In this example, you see that *User* has not been defined before, and is used as an actor.

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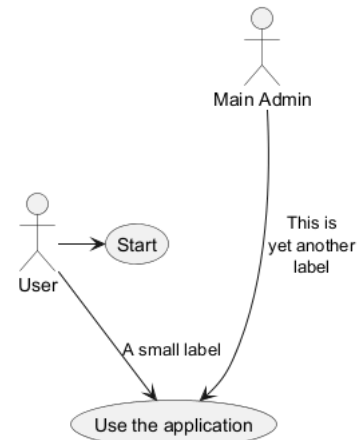
Sequence Use Case Class Activity Component State Object Deployment Tir



```
User -> (Start)
User --> (Use the application) : A small label

:Main Admin: ---> (Use the application) : This is\nyet another\nlabel

@enduml
```



Extension

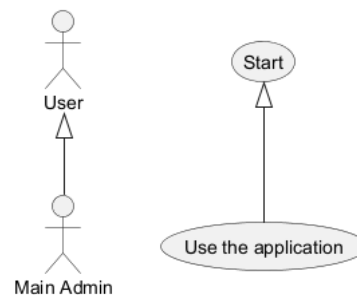
If one actor/use case extends another one, you can use the symbol <|-- .



```
@startuml
:Main Admin: as Admin
(Use the application) as (Use)

User <|-- Admin
(Start) <|-- (Use)

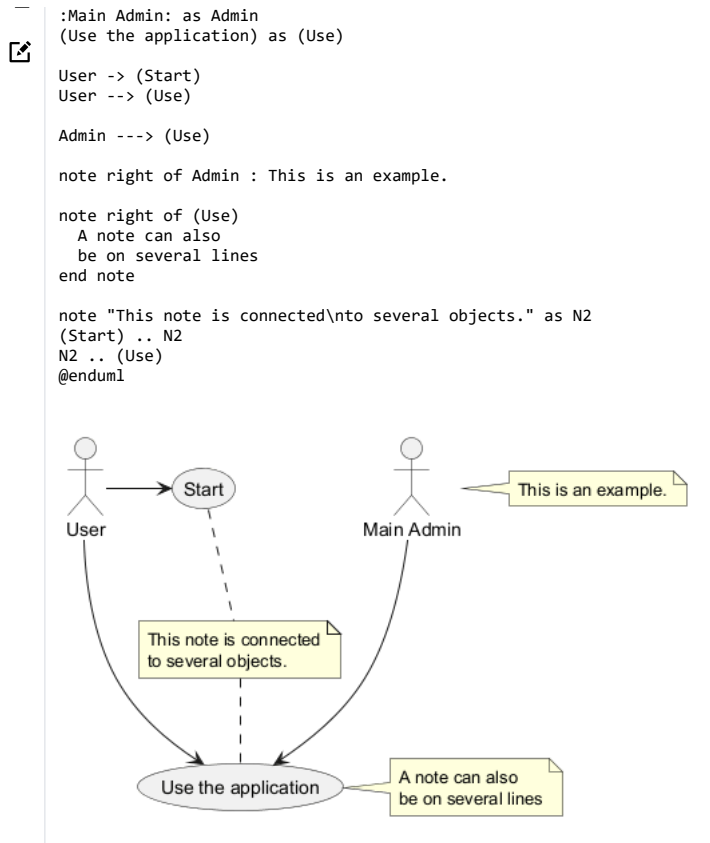
@enduml
```



Using notes

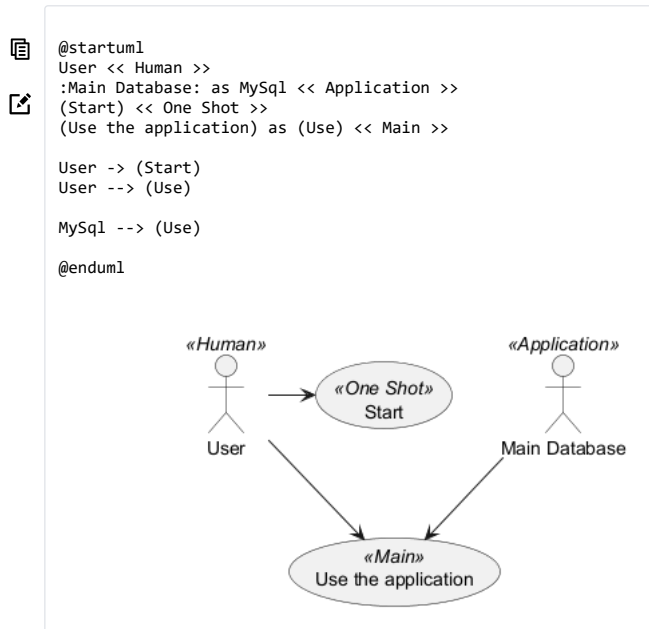
You can use the note left of , note right of , note top of , note bottom of keywords to define notes related to a single object.

A note can be also define alone with the note keywords, then linked to other objects using the .. symbol.



Stereotypes

You can add stereotypes while defining actors and use cases using << and >> .



Changing arrows direction

By default, links between classes have two dashes -- and are vertically oriented. It is possible to use horizontal link by putting a single dash (or dot) like this:

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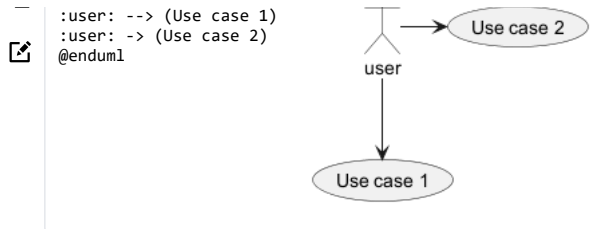
Preprocessing

Standard Library

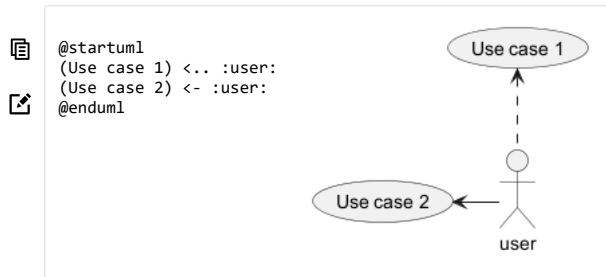
PDF Guide

Hitchhiker's Guide

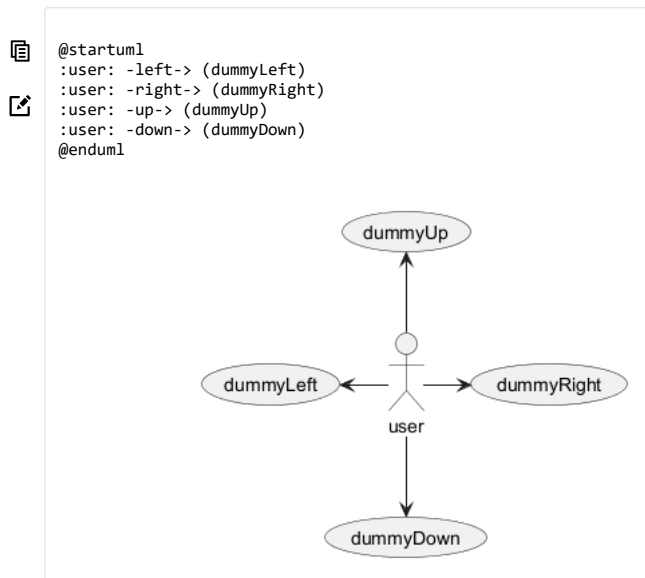
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You can also change directions by reversing the link:



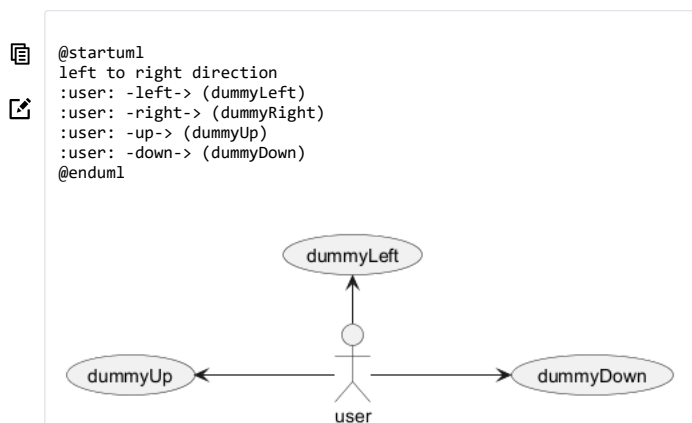
It is also possible to change arrow direction by adding `left`, `right`, `up` or `down` keywords inside the arrow:



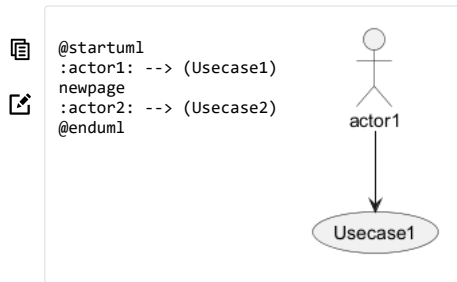
You can shorten the arrow by using only the first character of the direction (for example, `-d-` instead of `-down-`) or the two first characters (`-do-`).

Please note that you should not abuse this functionality : *Graphviz* gives usually good results without tweaking.

And with the `left to right direction` parameter:

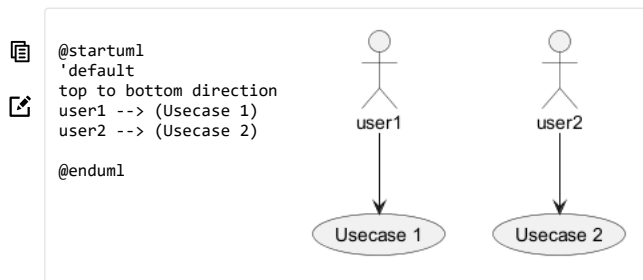


The `newpage` keywords to split your diagram into several pages or images.

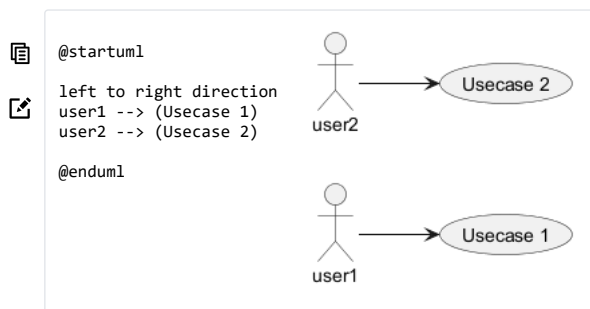


Left to right direction

The general default behavior when building diagram is **top to bottom**.



You may change to **left to right** using the `left to right direction` command. The result is often better with this direction.



See also 'Change diagram orientation' on [Deployment diagram](#) page.

Skinparam

You can use the `skinparam` command to change colors and fonts for the drawing.

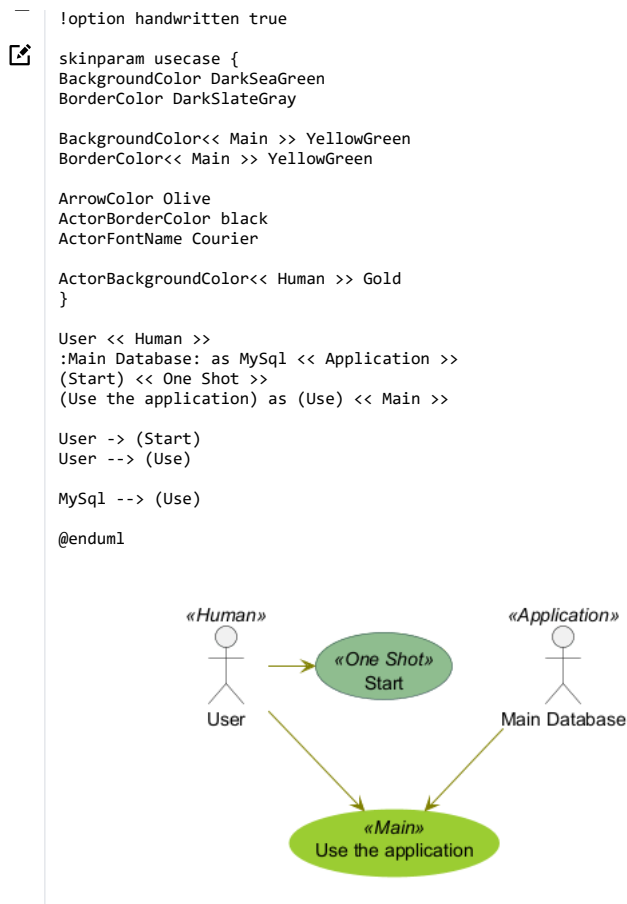
You can use this command :

- In the diagram definition, like any other commands,
- In an [included file](#),
- In a configuration file, provided in [the command line](#) or [the ANT task](#).

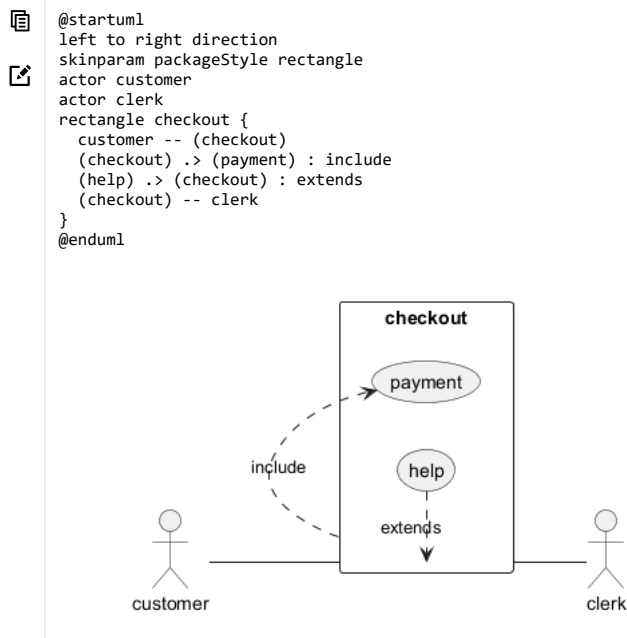
You can define specific color and fonts for stereotyped actors and usecases.

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Complete example



Business Use Case

You can add / to make Business Use Case.

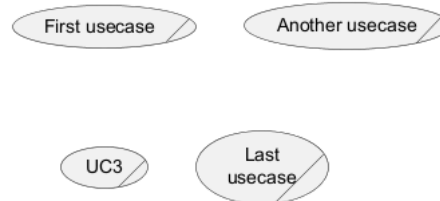
Business Usecase

```

(First usecase)/
(Another usecase)/ as (UC2)
usecase/ UC3
usecase/ (Last\nusecase) as UC4

@enduml

```



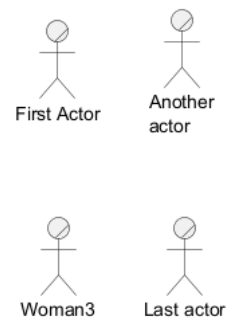
Business Actor

```

@startuml
:First Actor:/
:Another\nactor:/ as Man2
actor/ Woman3
actor/ :Last actor: as Person1

@enduml

```



[Ref. [QA-12179](#)]

Change arrow color and style (inline style)

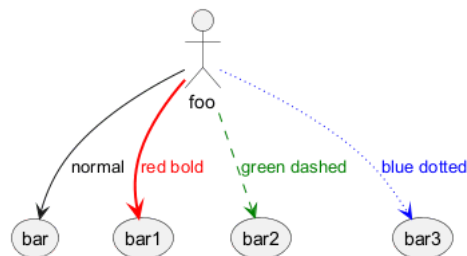
You can change the [color](#) or style of individual arrows using the inline following notation:

- #color;line.[bold|dashed|dotted];text:color

```

@startuml
actor foo
foo --> (bar) : normal
foo --> (bar1) #line:red;line.bold;text:red : red bold
foo --> (bar2) #green;line.dashed;text:green : green dashed
foo --> (bar3) #blue;line.dotted;text:blue : blue dotted
@enduml

```



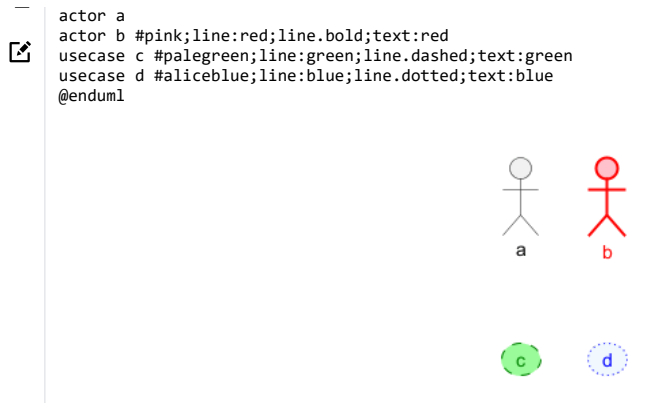
[Ref. [QA-3770](#) and [QA-3816](#)] [See similar feature on [deployment-diagram](#) or [class diagram](#)]

Change element color and style (inline style)

You can change the [color](#) or style of individual element using the

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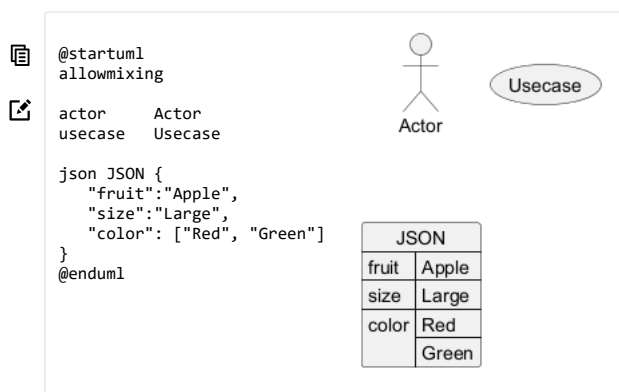
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[Ref. [QA-5340](#) and adapted from [QA-6852](#)]

Display JSON Data on Usecase diagram

Simple example



[Ref. [QA-15481](#)]

For another example, see on [JSON page](#).

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